Is a wine's score about the wine or the critic? Hypothesis tests on 120k wine reviews

Group name	Data Brokers
Dataset	kaggle.com/datasets/zynicide/wine-reviews
Code	github.com/hhiten10/DATA_BROKERS_DSC (notebook and README)

#	Name	Roll no.	Institute email	Contribution
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1. Dataset

Source: Kaggle, "Wine Reviews" by zynicide, downloaded on 22 October 2026. The reviews come from WineEnthusiast magazine and were collected in 2017. We use only `winemag-data-130k-v2.csv`. Each row is one review of one wine.

Problems in the raw data: repeated rows, missing prices and critic names, the vintage year hidden inside the wine's name, extra spaces in the text, the same grape under two names (Syrah and Shiraz), and a few very expensive bottles (up to $3,300).

	Raw file	After encoding
Instances (rows)	129,971	119,895
Attributes (columns)	13	26
Instances × attributes	1,689,623	3,117,270

Size check: $119{,}895 \times 26 = 3{,}117{,}270 \geq 3{,}000{,}000$, so the requirement is met. The 26 columns are 5 number columns (`points`, `price`, `price_imputed`, `vintage`, `desc_words`) plus one-hot columns for `country` (10) and `variety` (11). If we had deleted the rows with a missing price, we would have only $111{,}511 \times 26 = 2{,}899{,}286$, which is below 3M.

Key features

Feature	Type	Real-world meaning
`points`	discrete	score from 80 to 100; what we study
`price`	continuous	price of the bottle in US dollars
`country`	nominal	country of the wine; 43 values
`province`	nominal	state or province, e.g. Washington; 425 values
`variety`	nominal	grape type; 707 values
`taster_name`	nominal	the critic who scored the wine; 19 critics
`title`	text	name of the wine; contains the vintage year
`description`	text	the critic's review; we count its words (`desc_words`)

2. Data preparation

Issue found	What we did	Why
9,983 rows are exact copies	removed them	the same review counted twice makes the sample look bigger than it is
33 reviews appear on two different wines	kept only the first	the same text should not count twice in Test 2
59 rows have no country, 1 has no variety	removed them	too few to matter, and we cannot guess them
8,384 prices are missing, more in some countries (20.0% in France, 0.4% in the US)	filled with the median price of the same country; marked in `price_imputed`; the tests skip these rows	median, because a few very expensive bottles pull the mean up; we keep the rows because their scores are real
1 in 5 reviews has no critic name	filled with `Unknown`	the review and score are still real; Tests 4–5 use only two named critics

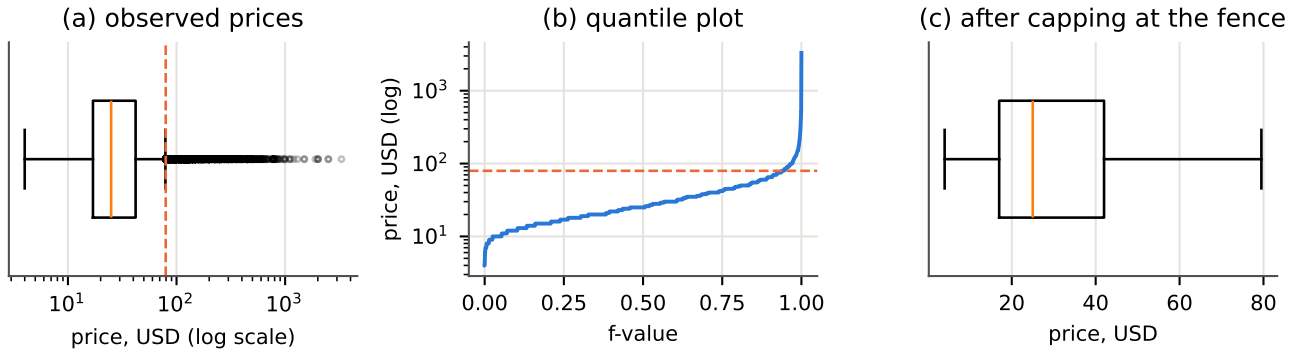


Figure 1: Prices have a long tail to the right: 6,867 bottles are above the $79.5 fence (dashed line) and the most expensive costs $3,300 (a, b). After capping at the fence, the long tail is gone (c).

Issue found	What we did	Why
4 columns not useful: `region_2` (61% missing), the Twitter handle (repeats the critic), `winery`, `designation` (too many values)	removed them	too empty, repeated, or too many values to compare groups
extra spaces in titles; a hidden space character in critic names	cleaned them	otherwise a search for a critic's name misses some rows
the vintage year is inside the wine's name	took it out with a regex; left empty if there is no year, it says NV, or there are two years	a wrong year is worse than an empty one
one grape, two names (e.g. Shiraz and Syrah)	merged 3 such pairs; 2,077 rows renamed	otherwise one grape is split into two groups
6,867 prices above the IQR fence $Q_3 + 1.5\,\mathrm{IQR} = \$79.5$ (Figure 1)	capped them at $79.5; the original price is kept in `price_raw`	real luxury wines, not mistakes, so we keep them; capping stops them pulling averages up
`country` and `variety` are categories	one-hot encoded the top 9 countries and top 10 varieties; the rest became Other	numbering them 1, 2, 3 would invent an order; very small groups are unreliable
no column for review length	added `desc_words`, the number of words	needed for Test 2

3. What changed after preparation

Measure	Before	After
Rows	129,971	119,895
Columns	13	26 after encoding
Exact copies	9,983	0
Reviews used twice	33	0
Missing prices	8,996	0 (8,384 filled and marked)
Missing critic names	26,244	0 (`Unknown`)
Grape varieties	707	698
Prices above $79.5	7,241	0 (capped)
Price kurtosis	829.5	0.1
Vintage year	inside the name	a number for 115,529 rows

Before is the raw file, so its counts include the copied rows. We only removed rows that were copies or had no country or variety; everything else was fixed and kept. Capping removed the extreme prices: kurtosis fell from 829.5 to 0.1. The new `vintage` and `desc_words` columns made Tests 2, 4 and 5 possible.

4. Basic questions and problem statement

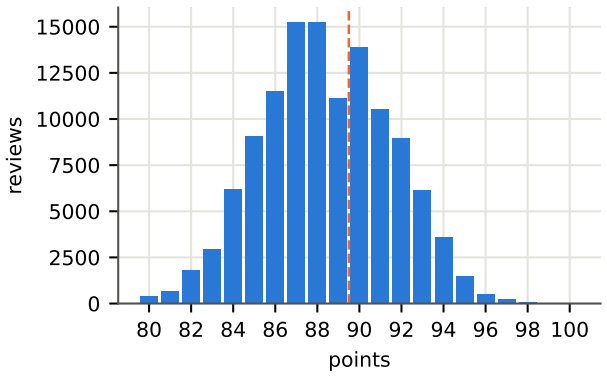


Figure 2: Scores go from 80 to 100 (mean 88.4, SD 3.1). There are none below 80 because the magazine does not publish them. 38% of wines score 90 or more.

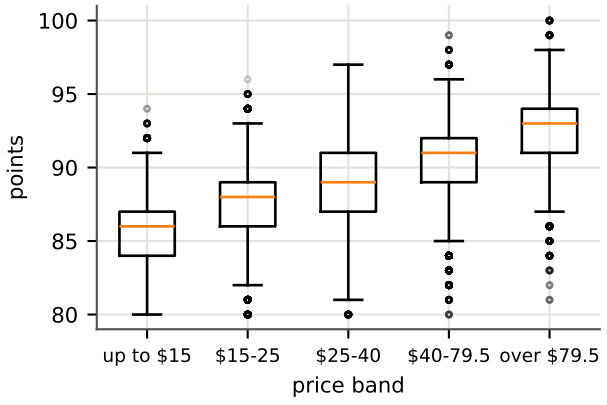


Figure 3: Pricier wines score higher: the median score rises from 86 to 93 across the price bands (correlation of log price with score: 0.62). So we must compare wines at the same price.

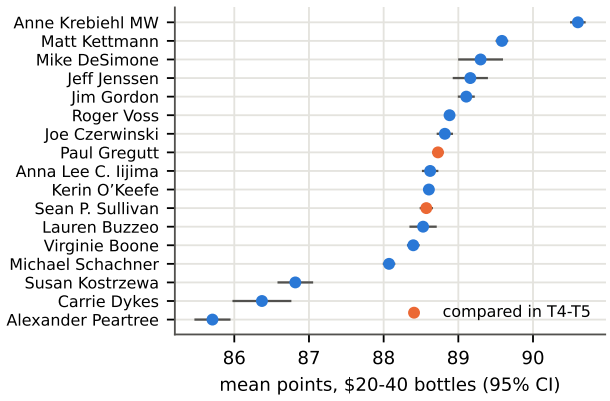


Figure 4: For $20–40 wines, the critics' average scores differ by up to 4.9 points, so the critic may matter as much as the wine. Orange: the two critics compared in Tests 4–5.

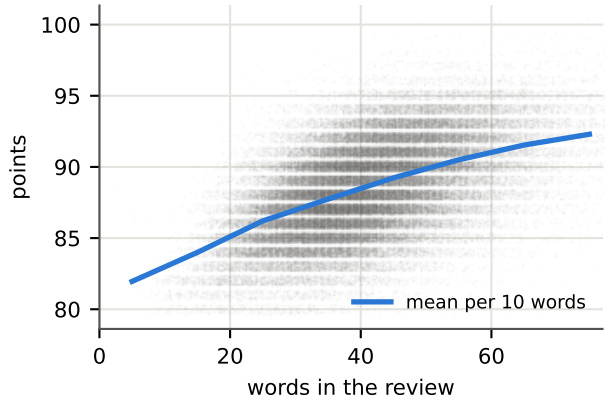


Figure 5: Longer reviews come with higher scores (correlation 0.54): about 82 points for the shortest reviews and 92 for the longest.

Basic questions

1. Do pricier wines score higher? Yes: the correlation of log price with score is 0.62.
2. Which countries score best? Austria (90.22 on average) and Germany (89.88), among countries with 500+ priced reviews.
3. Which grapes are reviewed most? Pinot Noir (12,274), Chardonnay (10,864) and Cabernet Sauvignon (8,835).

Our problem statements

P1 (main). For the same kind of wine, does the critic change the score? We check how often two critics give 90+ (Test 4) and how spread out their scores are (Test 5).

P2. At the same price, do Old-World wines score differently from New-World wines? (Test 1)

P3. Do longer reviews go with higher scores? (Test 2)

P4. At what price does a 90+ score become likely? (Test 3)

How they are new: the basic questions look at one column at a time. Our questions compare like with like (same price, or same state and years) and need columns we made ourselves (price band, Old/New World, vintage, word count).

5. Statistical inferences

Rows tested: the 111,511 reviews with a real price; the 8,384 filled-in prices are left out. We use $\alpha = 0.05$ and 95% confidence intervals (CIs). With this many rows, even a tiny difference is significant. So before each test we decide the smallest difference that would matter in real life, and after the test we compare the CI with it: *matters* if the whole CI is beyond it, *does not matter* if the whole CI is inside it, and *unclear* if the CI crosses it.

Test 1 (P2): Old-World vs New-World wines at $20–40

a) Hypotheses, written before running the test

H_0	$\mu_{\text{Old}} - \mu_{\text{New}} = 0$
H_1	$\mu_{\text{Old}} - \mu_{\text{New}} \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	two-sample t-test (an F-test first checks if the variances are equal)
Smallest difference that matters	1 point

$\mu_{\text{Old}}, \mu_{\text{New}}$: average score of $20–40 wines from the Old World (Europe and nearby countries such as Israel and Georgia) and the New World (the Americas, Australia, New Zealand, South Africa). Two-sided, because we did not guess a direction. 1 point, because scores are whole numbers, so a smaller gap would not change a printed score. Groups: 18,047 Old-World and 27,929 New-World wines.

b) Other tests we considered

Two separate groups, a score that is a number, and an unknown σ: so a two-sample t-test. The F-test says the variances are not equal ($F = 0.765$, $p = 7.94 \times 10^{-86}$), so we use Welch's t-test, which does not need equal variances.

Also considered	Why we did not use it
Pooled t-test	assumes equal variances, which is not true here. We ran it anyway: $t = 23.95$, same decision
Two-sample z-test	needs the true σ, which we do not know
Paired t-test	no wine is in both groups, so there is nothing to pair
χ^2 test	would only use 90+ yes/no and lose the exact score

c) Result

Estimate	$+0.595$ point (88.828 Old World, 88.232 New World)
95% confidence interval	0.548 to 0.643
Test statistic (d.f.)	$t = 24.64$ (42,033)
p-value	3.84×10^{-133}
Decision	reject H_0

Old-World wines score a little higher, and the difference is real. But the whole CI is below 1 point, so it

does not matter in practice.

Test 2 (P3): longer reviews vs higher scores

a) Hypotheses, written before running the test

H_0	$\rho \leq 0$
H_1	$\rho > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	t-test on the correlation r, df $= n - 2$
Smallest effect that matters	$\rho = 0.3$

ρ: correlation between the number of words in a review and its score. One-sided, because P3 asks about *higher* scores. 0.3, because a smaller correlation explains less than 9% of the score ($0.3^2 = 0.09$).

b) Other tests we considered

Both are numbers measured on the same review, so we test their correlation with a t-test.

Also considered	Why we did not use it
Welch t-test, long vs short reviews	a rougher question (reviews split at the median length). We ran it: long reviews score 2.694 points higher (CI 2.661 to 2.727), same decision
χ^2 test	would need us to cut review length into groups
Paired t-test	nothing to pair

c) Result

Estimate	$r = 0.538$
95% confidence interval	0.534 to 0.542
Test statistic (d.f.)	$t = 213.07$ (111,509)
p-value	≈ 0
Decision	reject H_0

The whole CI is above 0.3, so it **matters**: longer reviews go with higher scores. This does not mean long reviews cause high scores; critics probably write more about wines they like.

Test 3 (P4): price band vs the chance of a 90+ score

a) Hypotheses, written before running the test

H_0	the chance of 90+ is the same in every price band
H_1	the chance of 90+ differs between bands
Significance level	$\alpha = 0.05$
Side	one-sided (a χ^2 test uses the upper tail)
Test	χ^2 test for independence, df $= (5-1)(2-1) = 4$
Smallest difference that matters	5 percentage points between neighbouring bands

Price bands: up to \$15, \$15–25, \$25–40, \$40–79.5 and over \$79.5. 5 percentage points (pp) means one more 90+ wine in every 20.

b) Other tests we considered

Price band and 90+ (yes/no) are both categories, so we use the χ^2 test for independence. Every expected count is at least 2,592, well above 5. A χ^2 test does not give one estimate with a CI, so part c) shows the 90+ share of each band with its CI.

Also considered	Why we did not use it
z-test for two proportions	compares only two bands at a time. We ran it on the cheapest vs the most expensive band: $+80.7$ pp (CI 79.8 to 81.6)
χ^2 goodness of fit	needs an expected distribution to compare with, and we do not have one

c) Result

$\chi^2 = 30{,}191.9$, df $= 4$, $p \approx 0$: reject H_0.

Price band	n	90+ share, % (CI)	Step up, pp (CI)
up to \$15	23,256	5.1 (4.8 to 5.4)	–
\$15–25	33,129	23.4 (23.0 to 23.9)	$+18.3$ (17.8 to 18.8)
\$25–40	25,984	45.7 (45.1 to 46.3)	$+22.2$ (21.5 to 23.0)
\$40–79.5	22,275	69.1 (68.5 to 69.7)	$+23.4$ (22.6 to 24.3)
over \$79.5	6,867	85.8 (85.0 to 86.6)	$+16.7$ (15.7 to 17.7)

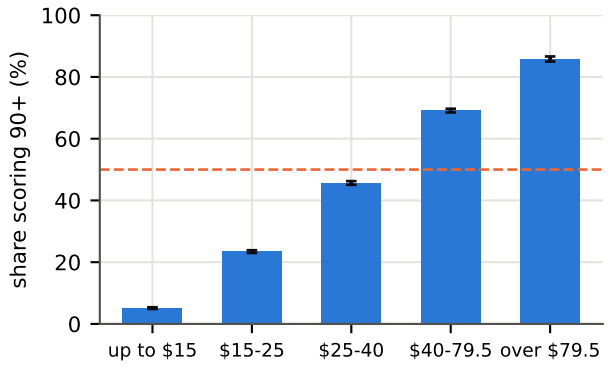


Figure 6: The chance of a 90+ score rises from 5% to 86% and passes one half (dashed line) in the \$40–79.5 band.

Each step up in price adds more than 5 pp, and every CI is above 5 pp, so all 4 steps **matter**. A 90+ score becomes more likely than not in the \$40–79.5 band.

Test 4 (P1): two critics, same kind of wine: how often 90+?

Paul Gregutt and Sean P. Sullivan both review wines from Washington State. To compare like with like, we use only Washington wines at \$20–40 from the years both critics reviewed at least 25 times (2009–2012, Figure 7). Gregutt scored 1,047 of these wines and Sullivan 806.

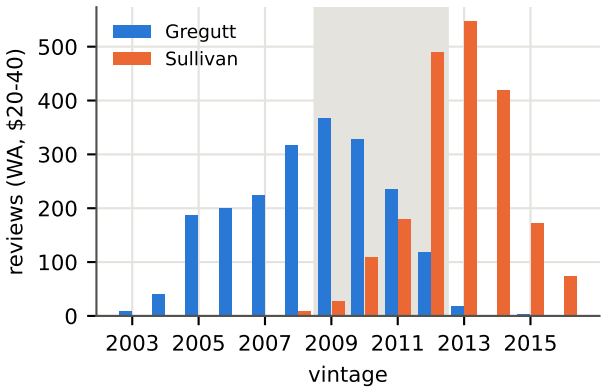


Figure 7: The two critics reviewed Washington wines in different years. Only 2009–2012 (shaded) has 25+ wines from both, so Tests 4 and 5 use only those years.

a) Hypotheses, written before running the test

H_0	$p_G - p_S = 0$
H_1	$p_G - p_S \neq 0$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	z-test for two proportions
Smallest difference that matters	5 percentage points

p_G, p_S: share of wines scored 90+ by Gregutt and by Sullivan. Two-sided, because we did not guess which critic is more generous. 5 pp means one more 90+ wine in every 20.

b) Other tests we considered

Two separate groups with a yes/no result (90+ or not), so we use the z-test for two proportions. Each group has well over 10 wines on each side of 90, so the normal approximation is fine.

Also considered	Why we did not use it
χ^2 test on the 2×2 table	gives the same answer ($\chi^2 = 24.16 = z^2$) but no CI for the difference
Welch t-test	compares average scores, which is a different question
Paired t-test	needs the same wines scored by both critics; only 1 of 1,851 wines was

c) Result

Estimate	$+11.4$ pp (Gregutt 49.3%, Sullivan 37.8%)
95% confidence interval	6.9 to 16.0 pp
Test statistic (d.f.)	$z = 4.92$ (no d.f. for z)
p-value	8.85×10^{-7}
Decision	reject H_0

For the same kind of wine, Gregutt gives 90+ more often. The whole CI is above 5 pp, so the critic **matters**. The years cannot explain it, because both critics are compared on the same years.

Test 5 (P1): two critics, same kind of wine: how spread out are the scores?

Same wines as Test 4.

a) Hypotheses, written before running the test

H_0	$\sigma_G^2/\sigma_S^2 = 1$
H_1	$\sigma_G^2/\sigma_S^2 \neq 1$
Significance level	$\alpha = 0.05$
Side	two-sided
Test	F-test for two variances
Smallest difference that matters	SD ratio of 1.2

σ_G^2, σ_S^2: variance of each critic's scores. An SD ratio of 1.2 means one critic's scores are 20% more spread out (about half a point), enough to move some wines across 90.

b) Other tests we considered

We compare spread, not averages, for two separate groups, so we use the F-test for two variances. It needs roughly normal scores; the Q-Q plots (Figure 8) show they are close to normal.

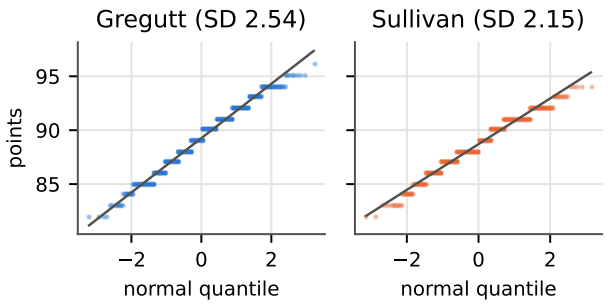


Figure 8: Both critics' scores are close to the normal line, but Sullivan's are less spread out (SD 2.15) than Gregutt's (SD 2.54).

Also considered	Why we did not use it
χ^2 test for one variance	treats Gregutt's variance as exactly known, which it is not, so its p-value is too small. We ran it: $\chi^2 = 575.60$, $p = 2.07 \times 10^{-10}$
z- and t-tests	compare averages, not spread

c) Result

Estimate	variance ratio 1.399 (SD ratio 1.183)
95% confidence interval	variance ratio 1.227 to 1.592; SD ratio 1.108 to 1.262
Test statistic (d.f.)	$F = 1.40$ (1,046, 805)
p-value	5.54×10^{-7}
Decision	reject H_0

Gregutt's scores are more spread out, and this is significant. But the CI for the SD ratio (1.108 to 1.262) crosses 1.2, so it is **unclear** whether the difference matters.

Summary of the five tests

	Question	Estimate (95% CI)	Matters if	Verdict
T1	Old vs New World, $20–40	$+0.595$ point (0.548 to 0.643)	1 point	does not matter
T2	review length vs score	$r = 0.538$ (0.534 to 0.542)	$r = 0.3$	matters
T3	price band vs 90+	steps of $+16.7$ to $+23.4$ pp	5 pp per step	all 4 steps matter
T4	Gregutt vs Sullivan, 90+ share	$+11.4$ pp (6.9 to 16.0)	5 pp	matters
T5	Gregutt vs Sullivan, spread	SD ratio 1.183 (1.108 to 1.262)	SD ratio 1.2	unclear

All five tests reject H_0, but the CIs show which differences are big enough to matter. **Answer to P1:** yes, the critic matters. For the same kind of wine, Gregutt gives 90+ 11.4 pp more often than Sullivan (Test 4). His scores are also more spread out, but we cannot say if that matters (Test 5). Old vs New World makes no real difference at the same price (Test 1), while price makes a big difference (Test 3).

6. Limitations

- **Only scores of 80 and above.** The magazine does not publish lower scores, so averages look higher and spreads look smaller.
- **Rows are not fully independent.** Wines from the same winery, or scored by the same critic, are related, so our p-values are too small. That is why we rely on the CIs and the smallest difference that matters.
- **Correlation is not cause.** Longer reviews do not cause higher scores. In Test 1, different critics review different countries, so origin and critic are mixed up. Gregutt and Sullivan may also have tasted wines from different wineries.
- **Filled-in prices are left out.** Prices are missing more often in some countries (20.0% in France), so these countries count less in our tests.
- **Small slices.** Tests 4–5 use one state, one price band, four years and two critics. Other cases may be different.
- **Old data.** The reviews were collected in 2017.